



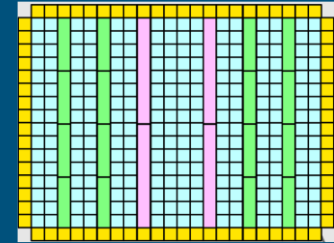
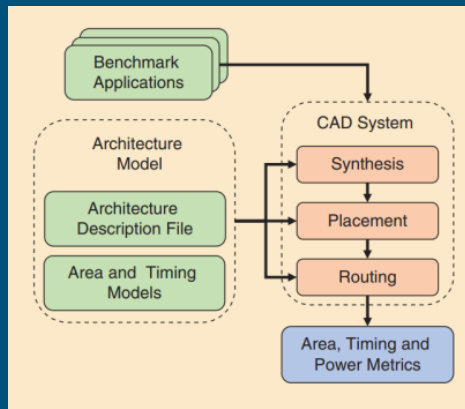
Lab 3 Supplement



Reconfigurable Computing
Spring 2026



Lab Objectives



Goal: FPGA Architecture and CAD Exploration using VTR

- Exploring FPGA architecture
- Exploring routing architecture
- Exploring CAD algorithms
- **Learn Tools:** Learn how to use the VTR framework
- **Main steps**
 - Compare FPGA logic block architectures
 - Compare FPGA routing architectures
 - Study effects of CAD parameters (placement & routing)

There is a lot of data collection, analysis and visualization in this lab. You can use LLMs to write scripts.

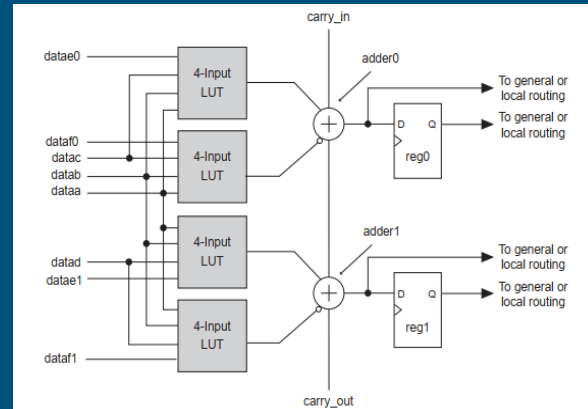
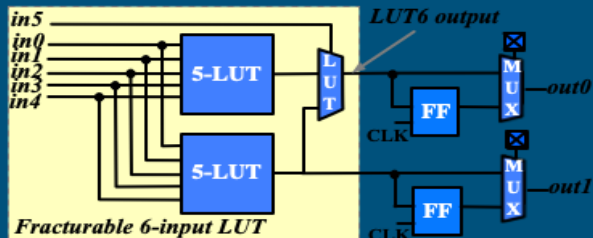
What you'll do in Lab 2

- Part 1 – Logic Block Architecture
 - Change CLB structure → measure metrics
- Part 2 – Routing Architecture
 - Change wire length & connectivity → measure metrics
- Part 3 – CAD Algorithms
 - Change placement & routing parameters → measure metrics
- VTR has been installed on SOL. What is VTR?
 - **VTR = Verilog-to-Routing**
 - It is a research framework consisting of:
 - **Synthesis** (Parmys)
 - **Logic optimization + tech mapping** (ABC)
 - **Packing, Placement & Routing** (VPR)

Part 1 - Logic Block Architecture

You will evaluate:

- **Arch1:** Basic 6-LUT architecture
- **Arch2:** Fracturable LUT + carry chain
- **Arch3:** Modified version of Arch1 ($N=5$, $I=20$, $O=5$)



What You Must Do in Part 1

- Copy Arch1 & Arch2
- Create Arch3
 - Change N = 5
 - Change I = 20
 - Change O = 5
 - Adjust area accordingly
- Answer architecture questions
- Configure a VTR task
- Run benchmarks
- Extract metrics: area & critical path delay

Part 2 – Routing Architecture

Architecture used: k6_N10_40nm.xml

You will evaluate variations of this architecture:

- **Different wirelengths**
- **Different block-to-routing connectivities**

What You Must Do in Part 2

- Run VPR
 - Find minimum channel width W
 - Re-run with $1.3W$
 - Measure routing area and critical path delay
- Experiment 1: Wire Length
 - Compare $L=1$, $L=3$, $L=16$
 - Measure and explain impact on routing area and critical path delay
- Experiment 2: F_c (Connection Box Flexibility)
 - Compare $F_c = 0.15, 0.5, 1.0$
 - Measure and explain impact on channel width, delay and area

Part 3 – CAD Algorithm Evaluation

Two studies:

- **Placement**
- **Routing**

What you must do in Part 3 Placement Study

- Run vpr on multiple benchmarks to observe the score changing with temperature
 - Plot the cost/score vs. temperature
- Run vpr with 5 different starting temperatures
 - Report mean and standard deviation of the cost/score with each starting temperature
- Change the placement algorithm to have greedy swaps
 - Discuss quality differences observed vs. normal results

What you must do in Part 3 Routing Study

- Run vpr on multiple benchmarks
 - Observe critical path delay and router runtime
- Change routing parameters and comment on the effect of parameters on critical path delay and router runtime
 - max_criticality
 - astar_fac
 - initial_pres_fac
 - etc.
- Disable A* routing and comment on the effect of parameters on critical path delay and router runtime

Extra Credit

- Early submission (up to 10%)
- Draw a figure showing the structure of the CLB in this architecture file. (3%)
- Evaluate the cost and benefit of partial local crossbar (3%)